

NovaPay: Cross-Border Digital Banking Platform

Product Requirements Document & Backend Architecture Specification

Product & Engineering Blueprint

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1. Executive Summary

NovaPay is a mobile-first, multi-currency financial platform designed for Africa-connected individuals, freelancers, and small businesses who need to hold, exchange, send, and spend money across borders. It is modeled on the product category pioneered by apps such as Eversend, Chipper Cash, and Wise, combining a multi-currency wallet, local and international payouts, virtual cards, and mobile-money integration into a single app.

The core value proposition is simple: let users hold several currencies in one account, convert between them at transparent rates, send money to bank accounts and mobile wallets across supported corridors, and spend internationally using a virtual card funded directly from a foreign-currency balance — without the hidden fees and card declines that come with traditional local bank accounts.

This document covers two things: the **product** (who it's for, what it does, how the experience is structured) and the **backend** (the services, data model, integrations, and infrastructure needed to build it safely and at scale).

2. Problem Statement & Target Users

Problem. People and small businesses across Africa (and the diaspora) routinely face: local cards being declined on foreign platforms (subscriptions, ad accounts, cloud services); banks charging high, opaque FX markups; slow, expensive traditional remittance channels; and no easy way to hold value in a stronger currency during local currency depreciation.

Primary user segments

Segment	Core need
Diaspora remittance senders	Fast, cheap transfers home to bank accounts / mobile

	money
Freelancers & remote workers	Receive USD/EUR from foreign clients, spend it online
Small business owners / importers	Pay foreign suppliers, hold multi-currency float
Students & travelers	Spend abroad without local card declines or FX markup
General savers	Hold USD/EUR or stablecoins as a hedge against local currency depreciation

3. Product Scope: Core Features

3.1 Multi-currency wallet

- A single account holds separate balances per currency (e.g., NGN, KES, UGX, GHS, USD, EUR, GBP).
- In-app conversion between held currencies at a visible mid-market rate plus a disclosed margin.
- Real-time balance and transaction history per currency.

3.2 Send money (payouts)

- **Bank payout** — send to a bank account in supported countries, resolved via local payment rails or partner aggregators.
- **Mobile money payout** — send directly to mobile wallets (e.g., M-Pesa, MTN MoMo, Airtel Money) in supported markets.
- **Wallet-to-wallet transfer** — instant, free transfers between users on the platform, identified by tag, phone number, or a shareable payment link.
- **International payout** — send to bank accounts abroad via SWIFT, SEPA, or local ACH-equivalent rails, or via stablecoin settlement where supported.

3.3 Receive money

- Dedicated USD and EUR virtual account numbers (via a banking-as-a-service partner) so users can receive ACH/SEPA/wire transfers directly into their NovaPay wallet — useful for freelancers being paid by foreign clients.
- Local collection accounts/mobile money numbers for receiving domestic payments.

3.4 Virtual & physical cards

- Instantly-issued virtual USD (and optionally EUR) Visa/Mastercard cards funded directly from the matching currency balance, avoiding double conversion.
- Card controls: freeze/unfreeze, single-use or merchant-locked cards, spend limits, card deletion and re-issuance.
- Physical card option for ATM withdrawal and in-person spend (phase 2).

3.5 Currency exchange

- Transparent FX conversion screen showing the live rate, the platform margin, and the exact amount the user will receive before they confirm.

3.6 Bill payments & airtime

- Airtime top-up and utility bill payments (electricity, TV, internet) in supported markets, as a retention and engagement feature.

3.7 Stablecoin rails (optional/advanced)

- Ability to receive and hold USDC/USDT/PYUSD, convert to local currency, or use to fund a card — used as a low-cost cross-border settlement rail in markets with thin USD liquidity.

3.8 Business accounts (phase 2)

- Multi-user business wallets with role-based approval limits, bulk payouts (payroll, supplier payments), and API access for merchants to accept and disburse payments.

3.9 Compliance & trust features

- KYC/KYB onboarding with document + selfie liveness verification.
- Transaction limits tied to verification tier.
- In-app dispute/chargeback flow and human support escalation.

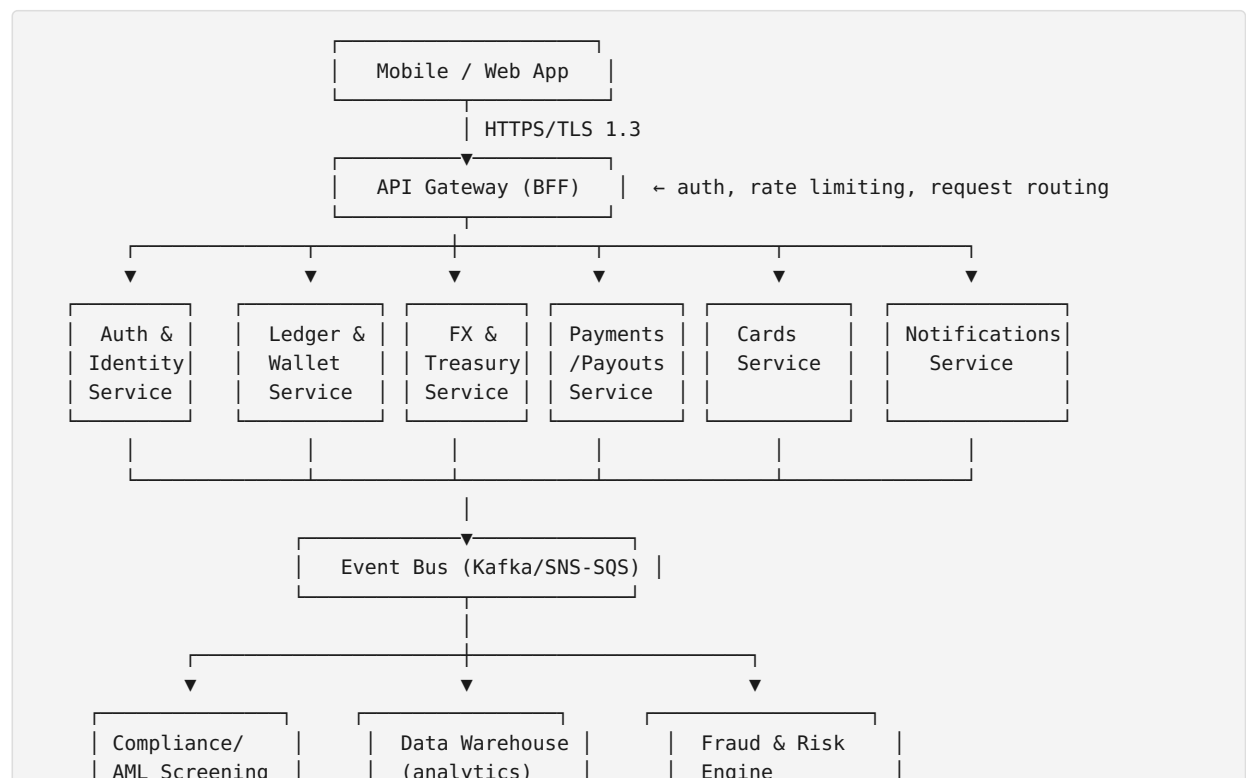
4. User Journeys (Illustrative)

1. **Sign up & verify** → phone/email OTP → BVN/NIN or passport + selfie liveness → tiered limits assigned.
2. **Fund wallet** → mobile money, local debit card, or bank transfer → funds land in local-currency balance.
3. **Convert** → user swaps NGN to USD at a shown rate → USD balance updates instantly.
4. **Spend** → user creates a virtual USD card → pays for a SaaS subscription abroad, funded from the USD balance, no second FX markup.
5. **Send home** → diaspora user in the UK funds a GBP balance, converts to KES, and pays out instantly to a family member's M-Pesa wallet.
6. **Receive as freelancer** → user requests a USD virtual account number, shares it with a US client, receives an ACH payment, converts part to local currency and keeps part in USD as savings.

5. Backend Architecture Overview

NovaPay's backend is best built as a set of independently deployable **microservices** behind an API gateway, rather than one monolith — this is standard for regulated fintech because it isolates the ledger and compliance-critical paths from feature surface area, and lets each service scale and be audited independently.

5.1 High-level architecture



5.2 Core services and responsibilities

Auth & Identity Service - User registration, login, OTP/2FA, session and device management. - KYC/KYB orchestration: document capture, liveness check, sanctions/PEP screening, verification tiering. - Integrates with third-party identity verification providers (e.g., Smile Identity, Youverify, Onfido) and national ID lookups (BVN/NIN in Nigeria, etc.).

Ledger & Wallet Service (the system of record) - Maintains the **double-entry ledger**: every movement of money is two balanced entries (debit + credit) across accounts, never a single mutable balance field. This is the single most important design decision in a financial backend — it guarantees the books always reconcile and gives a full, immutable audit trail. - Owns per-user, per-currency wallet balances, computed as the running sum of ledger entries, not stored as an editable number. - Exposes idempotent APIs for crediting/debiting balances, so retried requests never double-post a transaction.

FX & Treasury Service - Fetches live mid-market rates from a rate provider, applies the platform's margin, and locks a quoted rate for a short window (e.g., 30–60 seconds) at the moment a user confirms a conversion. - Manages the company's own currency float/nostro positions across partner banks, and rebalances liquidity between currencies.

Payments & Payouts Service - Abstracts over multiple payment rails and partners behind one internal interface, so the rest of the system doesn't care whether a payout went via a mobile-money aggregator, a local bank rail, or a SWIFT/SEPA partner. - Handles payout initiation, status polling/webhooks, retries, and failure reconciliation. - Talks to banking-as-a-service (BaaS) / payment partners such as regional mobile-money aggregators (e.g., a pan-African payments aggregator), a card-issuing processor (e.g., Marqeta, Union54, or a regional processor), and correspondent banks for USD/EUR virtual accounts and SWIFT/SEPA rails.

Cards Service - Talks to a card-issuing processor's API to create, freeze, top up, and terminate virtual (and physical) cards. - Enforces spend controls and syncs authorization/settlement webhooks back into the ledger in real time (an authorization should place a hold; settlement should convert the hold into a real ledger entry).

Fraud & Risk Engine - Real-time rules + ML scoring on signup, funding, and payout events (velocity checks, device fingerprinting, mismatched geolocation, sanctioned-country checks). - Feeds a manual review queue for flagged transactions.

Compliance/AML Service - Transaction monitoring for structuring, high-risk corridors, and sanctions list matches (OFAC, UN, local regulator lists). - Suspicious Activity Report (SAR) generation workflow and regulator reporting exports.

Notifications Service - Push, SMS, and email notifications for transaction events, security alerts, and marketing (with consent).

Support/Admin backend - Internal tools for KYC review, dispute handling, manual transaction adjustments (always via ledger-safe reversal entries, never direct balance edits), and customer support case management.

5.3 Data model (simplified core entities)

User	(id, phone, email, kyc_tier, status, created_at)
Wallet	(id, user_id, currency, status)
LedgerEntry	(id, wallet_id, amount, direction[debit credit], transaction_id, balance_after, created_at)
Transaction	(id, type[fund convert send payout card_spend], status, source_wallet_id, dest_wallet_id, fx_rate, fee, external_reference)
Card	(id, user_id, wallet_id, processor_ref, status, limits)
Beneficiary	(id, user_id, type[bank mobile_money], country, details)
KYCRecord	(id, user_id, document_type, verification_status, provider_ref)

The Transaction table is the human-readable event; the LedgerEntry rows are the accounting truth that must always net to zero across a transaction.

5.4 Key technical decisions

Concern	Recommended approach
Ledger integrity	Double-entry, append-only ledger; balances derived, never directly mutated
Idempotency	Every write API requires an idempotency key to prevent duplicate postings on retry
Consistency	Strong consistency (e.g., PostgreSQL with serializable transactions) for the ledger; eventual consistency acceptable for notifications/analytics
Service communication	REST/gRPC for synchronous calls, event bus (Kafka or SNS/SQS) for async events like “payout.completed”
Secrets & PCI scope	Card PANs and sensitive card data never touch NovaPay’s own servers — tokenized via the card processor; keep the platform out of full PCI-DSS Level 1 scope where possible
Rate limiting & abuse control	API gateway-level rate limits per user/IP, plus fraud engine scoring on every money-movement endpoint
Auditability	Every ledger mutation and admin action is immutably logged with actor, reason, and timestamp

5.5 Security & regulatory considerations

- Encryption in transit (TLS 1.3) and at rest (AES-256) for all PII and financial data.
- Role-based access control and least-privilege internal tooling; production database access gated and logged.
- Licensing: in most jurisdictions this requires either an Electronic Money Institution (EMI) license, a Payment Service Provider license, or a partnership with an already-licensed bank/EMI that provides the underlying regulated rails — this is a legal/regulatory dependency, not just an engineering one, and should be resolved with counsel per target market before launch.
- Data protection compliance (e.g., NDPR in Nigeria, GDPR for EU-resident users) for any personal data collected.

5.6 Third-party dependencies to plan for

- **Identity verification:** liveness/document verification provider.
- **Card issuing:** a card-issuing processor with African or global reach.
- **Banking-as-a-service:** for USD/EUR virtual accounts and SWIFT/SEPA connectivity.
- **Mobile money aggregator(s):** for payouts/collections across supported African markets.
- **FX rate feed:** a market-data provider for live mid-market rates.
- **Sanctions/PEP screening:** a compliance data provider.

6. Suggested Build Roadmap

Phase	Focus
Phase 0	Regulatory pathway, banking/BaaS partner selection, licensing strategy
Phase 1 (MVP)	Wallet + KYC + one funding method + one payout corridor + basic ledger
Phase 2	FX conversion, virtual USD card, additional payout

	corridors
Phase 3	USD/EUR receiving accounts, mobile money payouts, bill payments
Phase 4	Business accounts, stablecoin rails, physical cards, public API

7. Closing Notes

The hardest part of a system like this is rarely the UI — it's the ledger correctness, the partner integrations for money movement, and the regulatory licensing path. Any team building this should treat the double-entry ledger and compliance workflow as the foundation to get right first, before investing heavily in feature breadth.